Peer Response

by Farhad Karimov - Tuesday, 9 December 2025, 1:36 PM
Hi Husain,
Your evaluation of Abi's situation delivers a powerful ethical assessment through your focus on p-hacking and selective analysis which create scientific integrity problems (Andrade, 2021). The selection of favorable statistical tests from valid data violates fundamental research ethics because it breaks the core principles of research honesty. The requirement from ACM for computing professionals to maintain transparency while serving the public good supports your position (ACM, 2018).
Your discussion about legal accountability stands as a vital point in your argument. The UK and EU have established laws which protect consumers from false nutritional information that appears on food labels. The public safety and regulatory compliance requirements match Gotterbarn et al.'s (2018) professional standards for computing experts who need to evaluate the negative effects of their work.
Your analysis of professional responsibility stands out as a significant point. The responsibility to prevent others from misusing his research findings according to Resnik & Hosseini (2025) requires Abi to provide complete and transparent methodological information and full disclosure. The proposed actions you presented follow a logical sequence that matches professional ethical standards for handling situations.
Your argument about social trust stands as the most essential point in your discussion. The practice of hiding negative food safety evidence according to Baba & Esfandiari (2023) leads to a major decline in public trust toward scientific research. Your evaluation of the situation presents a complete and fair assessment of all ethical aspects.

References:

ACM (2018) *ACM Code of Ethics and Professional Conduct*.
Andrade, C. (2021) ‘HARKing, cherry-picking, p-hacking…’, *Journal of Clinical Psychiatry*, 82(1).
Baba, F.V. & Esfandiari, Z. (2023) ‘Risk communication in food safety’, *Heliyon*, 9(7).
Erasmus, A. (2025) ‘p-hacking: Its costs…’, *Erkenntnis*, 90(7).
Gotterbarn, D. et al. (2018) ‘ACM Code…’, *SIGUCCS Annual Conference*.
Legislation.gov.uk (2008) *Consumer Protection from Unfair Trading Regulations*.
Resnik, D.B. & Hosseini, M. (2025) ‘Ethics of AI in research’, *AI and Ethics*, 5(2).

Peer Response

by Abdulrahman Alhashmi - Saturday, 13 December 2025, 8:10 PM
Your post nails it: ethical risks go beyond fake data to include biased analysis that shapes the story. I agree. The real line is clear. Exploratory work counts as valid. But label it as such. Report it next to the main planned analysis.
Two steps would sharpen Abi's methods. They would curb misuse too.
First, fix the analysis plan fast.
Pre-register key outcomes. List model picks, exclusions, and p-value cut-offs.
This curbs researcher freedom. That freedom can tweak results to seem strong (Wicherts et al., 2016; Nosek et al., 2018). Big reviews prove the point. P-hacking shows up near standard cut-offs. It warps what seems solid (Head et al., 2015).
Second, report results tough to pick from.
Split them. Put primary first, exploratory last.
Show effect sizes and confidence intervals. Skip sole focus on p-values.
Share code that runs. Add a brief note for non-experts: "What this does not prove."
This meets standards for honest stats work (Royal Statistical Society, 2014).
On law and public effects, health claims by food makers face rules. They must rest on full proof, not slanted views (Regulation (EC) No 1924/2006). False food info risks UK law breaks too (Food Safety Act 1990, s.15). If Abi spots likely misuse, raise it with leaders. Alert a regulator if needed. That serves the public good.

References

Food Safety Act 1990 (UK), s.15.
Head, M.L. et al. (2015) ‘The Extent and Consequences of P-Hacking in Science’, PLOS Biology, 13(3).
Nosek, B.A. et al. (2018) ‘The preregistration revolution’, Proceedings of the National Academy of Sciences, 115(11).
Regulation (EC) No 1924/2006 on nutrition and health claims made on foods (2006).
Royal Statistical Society (2014) Code of Conduct.
Wicherts, J.M. et al. (2016) ‘Degrees of Freedom in Planning, Running, Analyzing, and Reporting Psychological Studies’, Frontiers in Psychology, 7.

Peer Response
by Jordan Speight - Monday, 15 December 2025, 5:10 AM

Your analysis of Abi’s dilemma is excellent, particularly your focus on the legal risks under consumer protection laws. I agree with you and Farhad that the ACM Code of Ethics (2018) obliges Abi to prioritise public welfare over client profit. However, I would like to expand on the definition of the problem you identified as "p-hacking".

While you frame the issue as intentional "hypothesis fishing," the ethical trap is often more subtle. Gelman and Loken (2013) describe the "garden of forking paths," where researchers make reasonable data-dependent decisions (e.g., excluding outliers, choosing control variables) that statistically invalidate the results even without malicious intent. This nuance is crucial because Abi might be facing not just pressure to lie, but pressure to exploit "researcher degrees of freedom" (Simmons et al., 2011) to find a "technically" defensible but misleading path.

Regarding the solution, while I support Abdulrahman’s suggestion of pre-registration, it cannot be enforced retroactively in a commercial contract. If Abi did not pre-register, he cannot simply claim he did. A more immediate practical step, aligned with the American Statistical Association’s statement on p-values (Wasserstein & Lazar, 2016), is to move the conversation away from binary "significance". Abi should perform a sensitivity analysis, demonstrating to the client that the positive result disappears under slight, reasonable changes to the model. This proves the findings are not robust enough for public claims without Abi needing to immediately accuse the client of fraud, effectively using the data to neutralise the pressure.

References

ACM. (2018). ACM Code of Ethics and Professional Conduct. Available at: **https://www.acm.org/code-of-ethics**

Gelman, A. and Loken, E. (2013) ‘The garden of forking paths: Why multiple comparisons can be a problem…’, *Department of Statistics, Columbia University*.

Simmons, J.P., Nelson, L.D. and Simonsohn, U. (2011) ‘False-Positive Psychology…’, *Psychological Science*, 22(11).

Wasserstein, R.L. and Lazar, N.A. (2016) ‘The ASA Statement on p-Values: Context, Process, and Purpose’, *The American Statistician*, 70(2).

Peer Response

by Rayyan Mohamed Abdalla Alshambeeli Alnaqbi - Thursday, 8 January 2026, 11:25 PM
Dear Abdulla,
Your conversation gives a full and thoughtful look at the moral, legal, and professional problems that Abi is facing. I agree with you that the issue goes beyond just data integrity and into the area of researcher intent. You point out that selective analysis, whether it's through p-hacking or confirmation bias, goes against the scientific method and makes it harder for researchers to find the truth (Andrade, 2021).
Your mention of professional codes of ethics makes your point even stronger. The ACM Code of Ethics clearly says that professionals must be honest, open, and produce high-quality work. Selective reporting is a direct violation of these rules (ACM, 2018). You also do a good job of linking these ethical breaches to legal consequences by pointing out that in places like the UK and EU, making false claims about food may break laws that protect consumers and require food to be labeled correctly.
I also agree with you that Abi is responsible for making sure that people don't misunderstand things. He can't stop people from misusing things downstream completely, but he can make sure that his findings are clear and raise concerns when they need to be. The steps you suggest—internal escalation, conditional reporting, and, if necessary, whistleblowing—are in line with widely accepted ethical best practices (Resnik & Hosseini, 2025).
In general, your post makes it clear that protecting public health and scientific integrity should come before business interests.

References:

- ACM (2018) *ACM Code of Ethics and Professional Conduct*. Available at: https://www.acm.org/code-of-ethics (Accessed: 20 Dec 2025).
- Andrade, C. (2021) 'Avoiding p-hacking and false-positive findings'. Available at: https://www.psychiatrist.com (Accessed: 20 Dec 2025).
- Resnik, D. and Hosseini, M. (2025) *Ethics in Research and Data Responsibility*. Available at: https://www.niehs.nih.gov/research/resources/bioethics (Accessed: 20 Dec 2025).